

Polylogarithmic Prime Divisors of Factorial Ratios and Vertical Binomial Coefficients

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Abstract

Let $F(N)$ be a fixed nontrivial balanced factorial ratio that is integral for every N , and let D be its height. Put $Y = (\log N / \log \log N)^{1/(1-\vartheta)}$. Assuming a quantitative lower bound for primes in intervals of length x^ϑ , where $1/2 < \vartheta < 1$, we prove that every sufficiently large N has $\gg_F Y / (\log \log N)^2$ distinct primes $p \ll_F Y$ satisfying $p^D \mid F(N)$; their logarithmic mass is $\gg_F Y / \log \log N$. The Baker–Harman–Pintz exponent $\vartheta = 21/40$ gives the cutoff $Y = (\log N / \log \log N)^{40/19}$. If the low-value set of the Landau step function is sufficiently small, a periodic-phase argument gives the optimal orders $Y / \log Y$ and Y . For every prescribed multiplicity $q \geq 1$, this applies to an explicit dyadic factorial ratio of height $2q + O(\sqrt{q})$. For each fixed R , the same prime-power construction yields two predetermined prime blocks depending on k and R , but not on n , such that, for every $2k \leq n \leq Rk$, one entire block divides $\binom{n}{k}$. The conclusion remains uniform for slowly expanding strips with $\log R = o(\sqrt{\log k})$, producing a pointwise superpolynomial squarefree part supported on subpolynomial primes.

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1 Introduction

Balanced factorial ratios

$$F(N) = \frac{\prod_{i=1}^r (a_i N)!}{\prod_{j=1}^s (b_j N)!}, \quad \sum_i a_i = \sum_j b_j,$$

form a natural meeting point of elementary, combinatorial, and analytic number theory. Landau’s criterion converts their integrality into the nonnegativity of a periodic step function. This viewpoint underlies both the classification theory of integral factorial ratios and structural bounds for their step functions [4, 5, 6, 7]. Here the question is different: how many small primes must divide a fixed integral ratio at a given argument, and how much squarefree mass must those primes carry?

Quantitative small-prime divisibility of central binomial coefficients goes back at least to Erdős, Graham, Ruzsa and Straus [8], who obtained average laws and a pointwise reciprocal-mass lower bound. Sander proved pointwise existence of prescribed prime-power divisors for near-central binomial and central multinomial coefficients [9, 10], while Kolesnik obtained exact prime-power divisibility for many large primes in multinomial settings [11]. These results use natural or polynomial prime ranges, large primes, special coefficient families, or averaging in the main parameter. The results below instead localize a growing set of distinct divisors below a polylogarithmic cutoff for

every sufficiently large argument, quantify its squarefree logarithmic mass, and apply to general balanced integral factorial ratios.

A least-prime-factor estimate supplies a single witness. The results here instead produce a *prime cloud*: a growing set of small divisors whose product is larger than every fixed power of the main parameter. Three forms of this phenomenon are proved.

1. Theorem 2.1 uses a terminal plateau of the Landau function and forces the maximal multiplicity allowed by the height. At the Baker–Harman–Pintz exponent, all supporting primes are $O_F((\log N / \log \log N)^{40/19})$.
2. Theorem 2.2 uses the full periodic phase rather than one plateau. When the low-value phase has sufficiently small measure, it recovers the optimal orders for both the number and logarithmic mass of the primes. Corollary 2.3 gives, for every prescribed q , an explicit dyadic ratio with many primes p satisfying $p^q \mid F(N)$.
3. Theorems 3.2 and 3.3 transfer the construction to Pascal’s triangle. For a prescribed strip, two prime blocks are chosen from k and the strip bound, independently of n ; for every n in that fixed or slowly expanding vertical ratio strip, one whole block divides $\binom{n}{k}$.

The common mechanism is to choose a prime-power exponent near $\log N / \log \log N$, place primes in short intervals around the associated root, and then multiplex many nearby exponents. The prime bands are disjoint, while Legendre’s or Lucas’s theorem converts their geometry into divisibility. The dense theorem adds periodic logarithmic averaging and a Brun–Titchmarsh deletion argument. Section 4 separates the factorial-ratio mechanism from the particular short-interval input, and Section 5 compares the resulting quantifiers and prime scales with nearby work.

Apart from standard background results (Landau’s criterion, Legendre’s formula, Lucas’s theorem, Hoeffding’s inequality, Brun–Titchmarsh, and usual prime-counting and Chebyshev upper bounds), the only deep analytic input is the stated quantitative short-interval theorem for primes. Exact finite certificates and dependency-free reproduction scripts accompany the source. They test the plateau mechanism on several factorial ratios and the two-block mechanism both exhaustively and on a genuine multi-prime example. These computations are supplementary and are not used in the asymptotic proofs.

Throughout, ϑ and the constants in QS_ϑ are fixed; implicit constants may depend on these data. For an integer M and $y \geq 2$, write

$$\omega_y(M) = \#\{p \leq y : p \mid M\}, \quad \text{rad}_{\leq y}(M) = \prod_{\substack{p \leq y \\ p \mid M}} p.$$

1.1 Quantitative short intervals

We use the following input, where $1/2 < \vartheta < 1$.

QS_ϑ . There are constants $c_0, X_0 > 0$ such that, for every $X \geq X_0$,

$$\pi(X) - \pi(X - X^\vartheta) \geq c_0 \frac{X^\vartheta}{\log X}.$$

Decreasing c_0 if necessary, we always assume $0 < c_0 \leq 1$.

Baker, Harman and Pintz prove in the final displayed estimate on p. 562 of [1] that

$$\pi(z + z^{21/40}) - \pi(z) \geq \frac{9}{100} \frac{z^{21/40}}{\log z} \tag{1.1}$$

for all sufficiently large z . Here and in (1.1), $\vartheta = 21/40$. Given a large X , choose the unique z with $z + z^\vartheta = X$. Then $z \sim X$, while $z = X - z^\vartheta > X - X^\vartheta$. Hence $[z, X] \subset [X - X^\vartheta, X]$, and (1.1) gives $QS_{21/40}$. Moreover,

$$\frac{z^{21/40}/\log z}{X^{21/40}/\log X} \longrightarrow 1.$$

Consequently $QS_{21/40}$ holds with every fixed $c_0 < 9/100$ after increasing X_0 if necessary. In all unconditional applications below we fix $c_0 = 0.08$.

We first record three tools used in both applications.

Lemma 1.1 (Translated prime blocks). *Assume QS_ϑ , put $h = x^\vartheta$, and fix real numbers $u < v$ with $v - u > 1$. Then*

$$\#\{p : x + uh \leq p \leq x + vh\} \gg_{u,v} \frac{h}{\log x}, \quad (1.2)$$

and

$$\sum_{x+uh \leq p \leq x+vh} \log p \gg_{u,v} h. \quad (1.3)$$

Proof. Apply QS_ϑ at $X = x + vh$. Since $X^\vartheta = (1 + o(1))h$, for all sufficiently large x one has

$$[X - X^\vartheta, X] \subset [x + uh, x + vh].$$

This proves (1.2). Every prime counted is $x(1 + o(1))$, so multiplying the count by $(1 + o(1)) \log x$ proves (1.3). $\square$

Lemma 1.2 (The optimized prime-power scale). *Fix $c > 0$ and a real parameter B_0 . Put*

$$d = 1 - \vartheta, \quad L = \log N, \quad \ell = \log L, \quad T = \ell - \log \ell + B_0,$$

and choose

$$m = \left\lfloor \frac{dL}{T} \right\rfloor, \quad x = (N/c)^{1/m}, \quad h = x^\vartheta. \quad (1.4)$$

Then, as $N \rightarrow \infty$,

$$\log x = \frac{T}{d} + o(1), \quad x^d = e^{B_0} \frac{L}{\ell} (1 + o(1)), \quad (1.5)$$

and

$$x \asymp_{B_0, c} \left(\frac{L}{\ell} \right)^{1/d}, \quad mx^{-d} = de^{-B_0} (1 + o(1)). \quad (1.6)$$

In particular, if $p = x + O(h)$, then

$$m \left| \log \frac{p}{x} \right| = O(mx^{-d}) = O(e^{-B_0}). \quad (1.7)$$

Proof. Since $T = O(\ell)$ and $m = dL/T + O(1)$,

$$\frac{1}{m} = \frac{T}{dL} \left(1 + O\left(\frac{T}{L}\right) \right).$$

As $\log(N/c) = L + O_c(1)$, this gives the first assertion in (1.5). Exponentiating $d \log x = T + o(1)$ gives the second. Finally,

$$mx^{-d} = \left(\frac{dL}{T} + O(1) \right) e^{-B_0} \frac{\ell}{L} (1 + o(1)) = de^{-B_0} (1 + o(1)),$$

because $T/\ell \rightarrow 1$. Since $h/x = x^{-d}$, (1.7) follows from $\log(1 + z) = O(z)$. $\square$

Lemma 1.3 (Multiplexing the prime-power exponent). *Retain the notation of Lemma 1.2, write m_0 for the integer in (1.4), and fix $\eta, V > 0$. Put*

$$J = \left\lfloor \eta \frac{m_0}{T} \right\rfloor, \quad \mathcal{M} = \{m_0 - J, m_0 - J + 1, \dots, m_0\}, \quad (1.8)$$

and, for $m \in \mathcal{M}$, define

$$x_m = (N/c)^{1/m}, \quad h_m = x_m^\vartheta. \quad (1.9)$$

Then

$$J \asymp \frac{L}{\ell^2}, \quad x_m \asymp x_{m_0} \asymp \left(\frac{L}{\ell} \right)^{1/d}, \quad \log x_m \asymp \ell \quad (1.10)$$

uniformly for $m \in \mathcal{M}$. Moreover,

$$mx_m^{-d} \leq m_0 x_{m_0}^{-d} = de^{-B_0}(1 + o(1)), \quad (1.11)$$

the intervals

$$[x_m, x_m + Vh_m] \quad (m \in \mathcal{M}) \quad (1.12)$$

are pairwise disjoint for all sufficiently large N , and

$$\sum_{m \in \mathcal{M}} \frac{h_m}{\log x_m} \asymp \frac{x_{m_0}}{\ell^2}, \quad \sum_{m \in \mathcal{M}} h_m \asymp \frac{x_{m_0}}{\ell}. \quad (1.13)$$

Proof. Put $A_0 = \log(N/c) = L + O_c(1)$. Since $J/m_0 \leq \eta/T = o(1)$,

$$0 \leq \log \frac{x_m}{x_{m_0}} = A_0 \left(\frac{1}{m} - \frac{1}{m_0} \right) = O_{\eta,d}(1),$$

which proves the uniform comparisons in (1.10); the estimate for J follows from $m_0 \asymp L/\ell$ and $T \asymp \ell$. Since $m \leq m_0$ and $x_m \geq x_{m_0}$, (1.11) is immediate from (1.6).

For consecutive exponents,

$$\begin{aligned} \frac{x_{m-1} - x_m}{h_m} &= x_m^d \left(\exp \left(\frac{A_0}{m(m-1)} \right) - 1 \right) \\ &\geq x_{m_0}^d \frac{A_0}{m_0(m_0-1)} \asymp_{B_0,d} \ell \longrightarrow \infty. \end{aligned} \quad (1.14)$$

Thus the right endpoint of the interval for m lies below x_{m-1} once N is large, proving (1.12). Finally, all h_m are comparable, so the left sides of (1.13) are respectively comparable to $Jx_{m_0}^\vartheta/\ell$ and $Jx_{m_0}^\vartheta$. Using $x_{m_0}^d \asymp L/\ell$ and $J \asymp L/\ell^2$ gives (1.13). $\square$

2 Integral factorial ratios

Let

$$F(N) = \frac{\prod_{i=1}^r (a_i N)!}{\prod_{j=1}^s (b_j N)!}, \quad (2.1)$$

where the two fixed tuples contain positive integers and

$$\sum_i a_i = \sum_j b_j. \quad (2.2)$$

Assume that common entries have been cancelled, that the ratio is nontrivial, and that $F(N)$ is an integer for every $N \geq 1$. Put

$$D = s - r, \quad A = \max(a_1, \dots, a_r, b_1, \dots, b_s).$$

Define Landau's step function by

$$\Delta_F(t) = \sum_i \lfloor a_i t \rfloor - \sum_j \lfloor b_j t \rfloor. \quad (2.3)$$

Theorem 2.1 (Factorial-ratio cloud). *Assume QS_ϑ . There are constants $c_F, C_F > 0$ such that, for every sufficiently large N ,*

$$\# \left\{ p \leq C_F \left(\frac{\log N}{\log \log N} \right)^{1/(1-\vartheta)} : v_p(F(N)) \geq D \right\} \geq c_F \frac{(\log N)^{1/(1-\vartheta)}}{(\log \log N)^{1/(1-\vartheta)+2}}. \quad (2.4)$$

Moreover, if $\mathcal{P}_N$ denotes the set on the left, then

$$\log \prod_{p \in \mathcal{P}_N} p \gg_F \frac{(\log N)^{1/(1-\vartheta)}}{(\log \log N)^{1/(1-\vartheta)+1}}. \quad (2.5)$$

For the published exponent $\vartheta = 21/40$, this reads

$$p \leq C_F \left(\frac{\log N}{\log \log N} \right)^{40/19}, \quad (2.6)$$

and

$$\#\mathcal{P}_N \gg_F \frac{(\log N)^{40/19}}{(\log \log N)^{78/19}}, \quad \log \prod_{p \in \mathcal{P}_N} p \gg_F \frac{(\log N)^{40/19}}{(\log \log N)^{59/19}}. \quad (2.7)$$

In particular, every $p \in \mathcal{P}_N$ occurs to exponent at least the height $D \geq 1$, and for every fixed $K > 0$,

$$\prod_{p \in \mathcal{P}_N} p > N^K \quad (2.8)$$

for all sufficiently large N .

Proof. Landau's criterion [4] says that $\Delta_F(t) \geq 0$ for all real t . Balance makes Δ_F one-periodic. On the universal terminal plateau

$$1 - \frac{1}{A} < t < 1,$$

every $\lfloor ct \rfloor = c - 1$, and therefore

$$\Delta_F(t) = s - r = D. \quad (2.9)$$

Thus $D \geq 0$. Also

$$\int_0^1 \Delta_F(t) dt = \frac{D}{2}.$$

If $D = 0$, nonnegativity forces $\Delta_F = 0$ almost everywhere. Both one-sided limits at each discontinuity are therefore zero. To spell out the resulting inversion, put

$$u_q = \#\{i : a_i = q\} - \#\{j : b_j = q\}.$$

The jump of Δ_F at $1/q$ is

$$J_q = \sum_{\substack{r \geq 1 \\ rq \leq A}} u_{rq} = 0. \quad (2.10)$$

Starting with $q = A$ and descending to $q = 2$ gives $u_q = 0$ at every step; then $\sum_q u_q = r - s = 0$ gives $u_1 = 0$. Thus the two coefficient multisets agree, contrary to the reduced nontrivial assumption. Hence $D \geq 1$.

At every continuity point,

$$\Delta_F(t) + \Delta_F(1 - t) = D. \quad (2.11)$$

Landau nonnegativity therefore gives $\Delta_F(t) \leq D$ there, and right continuity extends this bound to every t . Together with (2.9), this shows that the maximum of Δ_F is exactly D . Take an open interval I inside the terminal plateau in (2.9), fix $c \in I$, and choose $\rho > 0$ so that

$$[ce^{-\rho}, ce^{\rho}] \subset I. \quad (2.12)$$

Apply Lemma 1.3 with this c , a fixed $\eta > 0$, and $V = 4$. Choose B_0 so large that (1.11) gives, uniformly for $m \in \mathcal{M}$,

$$4mx_m^{\eta-1} < \rho \quad (2.13)$$

for all sufficiently large N . For each such m , set

$$\mathcal{Q}_{N,m} = \{p \text{ prime} : x_m + 2h_m \leq p \leq x_m + 4h_m\}. \quad (2.14)$$

Lemma 1.1 gives

$$|\mathcal{Q}_{N,m}| \gg \frac{h_m}{\log x_m}, \quad \sum_{p \in \mathcal{Q}_{N,m}} \log p \gg h_m. \quad (2.15)$$

These sets are pairwise disjoint by (1.12). If $p \in \mathcal{Q}_{N,m}$, then

$$\frac{N}{p^m} = c \exp\left(-m \log \frac{p}{x_m}\right), \quad 0 \leq m \log \frac{p}{x_m} \leq 4m \frac{h_m}{x_m} < \rho,$$

so $N/p^m \in I$. Landau nonnegativity and Legendre's formula now give

$$\begin{aligned} v_p(F(N)) &= \sum_{\nu \geq 1} \left(\sum_i \left\lfloor \frac{a_i N}{p^\nu} \right\rfloor - \sum_j \left\lfloor \frac{b_j N}{p^\nu} \right\rfloor \right) \\ &= \sum_{\nu \geq 1} \Delta_F\left(\frac{N}{p^\nu}\right) \geq \Delta_F\left(\frac{N}{p^m}\right) = D. \end{aligned} \quad (2.16)$$

Let $\mathcal{Q}_N = \bigcup_{m \in \mathcal{M}} \mathcal{Q}_{N,m}$. By (1.10), all its primes lie below $C_F(\log N / \log \log N)^{1/d}$; by (1.13) and (2.15),

$$|\mathcal{Q}_N| \gg_F \frac{x_{m_0}}{\ell^2}, \quad \sum_{p \in \mathcal{Q}_N} \log p \gg_F \frac{x_{m_0}}{\ell}. \quad (2.17)$$

Thus $\mathcal{Q}_N \subseteq \mathcal{P}_N$, and (1.10) translates (2.17) into (2.4)–(2.7). Finally,

$$\frac{x_{m_0}/\ell}{\log N} \asymp \frac{(\log N)^{1/d-1}}{(\log \log N)^{1/d+1}} \longrightarrow \infty,$$

which proves (2.8). □

Consequences

1. For every height $D \geq 1$, each sufficiently large term has a growing collection of polylogarithmic primes whose D -th powers all divide the term. Equivalently, it has a large perfect D -th-power divisor supported on polylogarithmic primes. When $D \geq 2$, this is much stronger than eventual nonsquarefreeness.
2. The conclusion survives polynomial-size normalization. More precisely, let $0 \neq Q \in \mathbb{Z}[X]$, and suppose $F(N)/Q(N)$ is integral whenever $Q(N) \neq 0$. Define the residual candidate set

$$\mathcal{Q}'_N = \{p \in \mathcal{Q}_N : p \nmid Q(N)\}.$$

Then

$$\sum_{p \in \mathcal{Q}_N \setminus \mathcal{Q}'_N} \log p \leq \log |Q(N)| = O_Q(\log N),$$

whereas the full logarithmic mass in (2.17) is $\omega(\log N)$. Moreover, since $\log p \asymp \log \log N$, the number of deleted candidates is

$$O_Q\left(\frac{\log N}{\log \log N}\right) = o\left(\frac{x_{m_0}}{(\log \log N)^2}\right)$$

since, in fact,

$$\frac{x_{m_0}/\ell^2}{L/\ell} \asymp \frac{(L/\ell)^{1/d-1}}{\ell^2} \rightarrow \infty.$$

Every residual prime satisfies $v_p(F(N)/Q(N)) = v_p(F(N)) \geq D$. Thus both the same-order count and the superpolynomial radical conclusion hold for $|F(N)/Q(N)|$ itself.

3. In particular, for every fixed $r \geq 1$, the Fuss–Catalan number

$$C_N^{(r)} = \frac{1}{rN+1} \binom{(r+1)N}{N}$$

has the same-order polylogarithmic prime cloud and superpolynomial small-prime radical. Indeed, all candidate primes satisfy $\log p \asymp \log \log N$, so a polynomial-size denominator removes only

$$O\left(\frac{\log N}{\log \log N}\right)$$

candidates. The displayed comparison in the preceding item shows that this is little-oh of the count in (2.4).

Structural restrictions on a dense cloud

For a fixed integer $q \geq 1$, put

$$\zeta_{F,q} = \text{meas}\{t \in [0,1) : \Delta_F(t) < q\}. \quad (2.18)$$

The mean-value identity used above gives

$$\zeta_{F,q} \geq 1 - \frac{D}{2q}. \quad (2.19)$$

There is a sharper integer obstruction. If $D \leq 2q - 1$, the reflection identity (2.11) shows that, for almost every pair $t, 1 - t$, at least one member has $\Delta_F < q$. Hence $\zeta_{F,q} \geq 1/2$. Since we have normalized $c_0 \leq 1$, the hypothesis below necessarily forces

$$D \geq 2q. \quad (2.20)$$

Moreover, $\Delta_F(t) = 0$ on $[0, 1/A)$, and therefore

$$\zeta_{F,q} \geq \frac{1}{A}, \quad \zeta_{F,q} < \frac{c_0 \vartheta}{2} \implies A > \frac{2}{c_0 \vartheta}. \quad (2.21)$$

For the BHP choice $c_0 = 0.08$, $\vartheta = 21/40$, the threshold is $c_0 \vartheta / 2 = 21/1000$, so necessarily $D \geq 2q$ and $A \geq 48$. In particular, the dense criterion can never apply to a height-one ratio.

Theorem 2.2 (A dense-cloud criterion). *Retain the explicit constant c_0 in QS_ϑ and the notation $\zeta_{F,q}$ from (2.18). If*

$$\zeta_{F,q} < \frac{c_0 \vartheta}{2}, \quad (2.22)$$

then there are constants $c_{F,q}, C_{F,q} > 0$ such that, for every sufficiently large integer N , on writing

$$Y = \left(\frac{\log N}{\log \log N} \right)^{1/(1-\vartheta)}, \quad (2.23)$$

$$\#\{p \leq C_{F,q} Y : v_p(F(N)) \geq q\} \geq c_{F,q} \frac{Y}{\log \log N},$$

and the logarithm of the product of these primes is at least $c_{F,q} Y$. Both orders are optimal for this cutoff, up to constants.

Before proving the theorem, we isolate the four estimates compressed into the bad-phase argument. In Auxiliary Lemmas Q, G, and B below, let $X \rightarrow \infty$, put $H = X^\vartheta$, and suppose the positive integer $m = m(X)$ has $mX^{\vartheta-1}$ in a fixed compact subinterval of $(0, \infty)$ and satisfies $\log H = o(m)$. For fixed $0 < \alpha < 1$, set $U = H^{\alpha/2}$, $V = H^\alpha$, the interval $I = [XV^{-1/m}, XU^{-1/m}]$, and $W = |I|$. For the fixed pair F, q , let $g(t) = 1_{\{\Delta_F(t) < q\}}$ be extended one-periodically, and put $\zeta = \int_0^1 g(t) dt = \zeta_{F,q}$.

Auxiliary Lemma (P: Periodic logarithmic mean). *If g is any bounded, Riemann-integrable, one-periodic function with mean equal to ζ , then the integral $\int_U^V g(t) dt/t$ differs from the main term $\zeta \log(V/U)$ by $O_g(U^{-1})$, uniformly whenever the bounds $1 \leq U < V$ hold.*

Proof. The function $G(y) = \int_0^y (g(t) - \zeta) dt$ is bounded and one-periodic. Integration by parts leaves the boundary term $[G(t)/t]_U^V$ and the residual integral $\int_U^V G(t)t^{-2} dt$; both are of order $O_g(U^{-1})$. $\square$

Auxiliary Lemma (Q: Quasi-short prime mass). *Assume QS_ϑ . Then the width satisfies the estimate $W \sim \alpha \vartheta X \log X / (2m) \asymp H \log X$. Moreover, the prime count $|\mathbb{P} \cap I|$ is at least the quantity $(c_0 - o(1))W / \log X$.*

Proof. Since $\log V = o(m)$, expansion of the two endpoints gives the width formula. Starting at $\sup I$, recursively take each half-open band of the form $(Z - Z^\vartheta, Z]$ while it fits inside I . Their scales are $X(1 + o(1))$, their lengths sum to $W - O(H)$, and the remainder has length $O(H) = o(W)$. The bands are disjoint, and $\pi(Z) - \pi(Z - Z^\vartheta)$ already counts the indicated half-open band. Summing QS_ϑ therefore proves the claim, with no boundary prime counted twice. $\square$

Auxiliary Lemma (G: Inverse-phase geometry). *Assume $\zeta < 1$. Define $E = \{y \in I : g(X^m/y^m) = 1\}$. Then the identity $|E| = (X/m) \int_U^V g(t)t^{-1-1/m} dt = (\zeta + o(1))W$ holds. Modulo endpoint conventions, E has $O_{F,q}(V)$ components. Every component whose phase interval is not cut by U or V has length bounded below by $c'_{F,q}H^{1-\alpha}$ for some $c'_{F,q} > 0$. The at most two cut components can be enlarged to complete phase components at a total length cost of order $O_{F,q}(H^{1-\alpha/2}) = o(W)$.*

Proof. The first identity is the change of variables $y = Xt^{-1/m}$. Uniformly on $[U, V]$, one has $t^{-1/m} = 1 + o(1)$; Auxiliary Lemma P and the width formula from Auxiliary Lemma Q give the measure asymptotic.

In one period the bad phase is, up to finitely many boundary points, a finite union of half-open intervals. If it is nonempty, its positive component lengths have a fixed minimum $\delta_{F,q} > 0$. The inverse map has derivative magnitude $(X/m)t^{-1-1/m}$, and $X/m \asymp H$. Thus a full inverse component has length at least the quantity $\delta_{F,q}(X/m)V^{-1-1/m} \gg_{F,q} H^{1-\alpha}$. The number of components is $O_{F,q}(V)$. Only those meeting U or V can be cut, and completing them costs at most $O_{F,q}((X/m)U^{-1-1/m}) = O_{F,q}(H^{1-\alpha/2})$. If $\zeta = 0$, the component assertions are vacuous apart from the same endpoint convention. $\square$

Auxiliary Lemma (B: Brun–Titchmarsh deletion and endpoint cleanup). *Under the hypotheses of Auxiliary Lemma G, let B_X denote the number of primes in E . Then the normalized count satisfies $\limsup_{X \rightarrow \infty} B_X \log X/W \leq 2\zeta/[\vartheta(1-\alpha)]$.*

Proof. Cover E by the complete component intervals from Auxiliary Lemma G. Their total length is $(\zeta + o(1))W$. Closing or extending real endpoints before applying Brun–Titchmarsh costs $O(1)$ per component, hence $O_{F,q}(V)$ in total. Every nonempty covering interval J has the lower bound $\log |J| \geq [\vartheta(1-\alpha) + o(1)] \log X$. Summing Brun–Titchmarsh over the components gives the estimate $B_X \leq (2\zeta/[\vartheta(1-\alpha)] + o(1))W/\log X + O_{F,q}(V)$. Since $V = H^\alpha = o(H)$ and $W/\log X \asymp H$, the endpoint term is absorbed into the limsup bound. $\square$

Proof. Write $L = \log N$, $\ell = \log L$, $d = 1 - \vartheta$, fix any constants $B_0 \in \mathbb{R}$ and $\eta > 0$, and choose the parameters $m_0, J, \mathcal{M}$ as in Lemma 1.3 with $c = 1$. Put

$$X_m = N^{1/m}, \quad H_m = X_m^\vartheta \quad (m \in \mathcal{M}).$$

Uniformly in this set,

$$X_m \asymp Y, \quad \log X_m \asymp \ell, \quad H_m \asymp \frac{Y\ell}{L}, \quad J \asymp \frac{L}{\ell^2}. \quad (2.24)$$

Moreover $mX_m^{\vartheta-1}$ stays in a fixed compact subinterval of the positive real axis, and $\log H_m = o(m)$, uniformly over the exponents $m \in \mathcal{M}$. Thus Auxiliary Lemmas Q, G, and B apply, every $o(1)$ below is uniform over this exponent set. Choose

$$0 < \alpha < 1 - \frac{2\zeta_{F,q}}{c_0\vartheta}, \quad (2.25)$$

and define

$$\begin{aligned} U_m &= H_m^{\alpha/2}, & V_m &= H_m^\alpha, \\ I_m &= \left[X_m V_m^{-1/m}, X_m U_m^{-1/m} \right]. \end{aligned} \quad (2.26)$$

Thus $p \in I_m$ exactly when $U_m \leq N/p^m \leq V_m$. If $W_m = |I_m|$, then

$$W_m = \left(\frac{\alpha\vartheta}{2} + o(1) \right) \frac{X_m \log X_m}{m} \asymp H_m \log X_m. \quad (2.27)$$

Auxiliary Lemma Q, including its half-open endpoint convention, also gives

$$\#(\mathbb{P} \cap I_m) \geq (c_0 - o(1)) \frac{W_m}{\log X_m}. \quad (2.28)$$

We next apply Auxiliary Lemma G to the bad phase. Let $g(t) = 1_{\{\Delta_F(t) < q\}}$, extended one-periodically. Under the change of variables $p = X_m t^{-1/m}$, the total length of the bad subset of I_m is

$$\frac{X_m}{m} \int_{U_m}^{V_m} g(t) t^{-1-1/m} dt = (\zeta_{F,q} + o(1)) W_m. \quad (2.29)$$

Indeed, $t^{-1/m} = 1 + o(1)$ uniformly here, and periodic summation gives

$$\int_U^V \frac{g(t)}{t} dt = \zeta_{F,q} \log \frac{V}{U} + O_{F,q}(U^{-1}).$$

The bad set in one period is a finite union of fixed intervals. Every full component of its inverse image in I_m has length $\gg_{F,q} H_m^{1-\alpha}$. Enlarging the two possible boundary components to full components adds only $O_{F,q}(H_m^{1-\alpha/2}) = o(W_m)$ total length. Applying the Brun–Titchmarsh inequality [2] to each enlarged component and using the phase-boundary endpoint cleanup from Auxiliary Lemma B gives

$$\#\{p \in I_m : \Delta_F(N/p^m) < q\} \leq \left(\frac{2\zeta_{F,q}}{\vartheta(1-\alpha)} + o(1) \right) \frac{W_m}{\log X_m}. \quad (2.30)$$

The choice (2.25) makes the quantity $c_0 - 2\zeta_{F,q}/[\vartheta(1-\alpha)]$ positive. The endpoint contribution absorbed in (2.30) is of size $O_{F,q}(V_m) = o(H_m)$ because $V_m = H_m^\alpha$ with $\alpha < 1$. By (2.25), subtracting (2.30) from (2.28) leaves $\gg_{F,q} H_m$ primes with $\Delta_F(N/p^m) \geq q$. Landau nonnegativity and Legendre’s formula then give $v_p(F(N)) \geq q$ for each one.

The intervals I_m are pairwise disjoint. In fact, putting $a = \alpha\vartheta < 1$, direct calculation gives

$$\log \inf I_{m-1} - \log \sup I_m = \frac{L((2-a)m^2 - (2+2a)m + a)}{2m^2(m-1)^2} > 0 \quad (2.31)$$

for all sufficiently large m . Summing over $m \in \mathcal{M}$ and using (2.24) yields

$$\sum_{m \in \mathcal{M}} H_m \asymp \frac{Y}{\ell}.$$

This proves (2.23). Since every resulting prime is comparable to Y , their logarithmic mass is $\gg_{F,q} Y$. Conversely, the standard prime-counting and Chebyshev upper bounds give only $O(Y/\log Y) = O(Y/\ell)$ primes below $C_{F,q}Y$, with total logarithmic mass $O(Y)$, proving order-optimality. $\square$

Corollary 2.3 (Optimal-order dyadic clouds with prescribed multiplicity). *For an integer $s \geq 1$, let*

$$\mathcal{F}_s(N) = \prod_{r=0}^{s-1} \binom{2^{r+1}N}{2^r N} = \frac{(2^s N)!}{(N!)^2 \prod_{r=1}^{s-1} (2^r N)!}. \quad (2.32)$$

Fix $1 \leq q \leq s$. If

$$2^{-s} \sum_{j=0}^{q-1} \binom{s}{j} < \frac{21}{1000}, \quad (2.33)$$

then there is a constant $C_{s,q} > 0$ such that, for every sufficiently large integer N , on writing

$$Y = \left(\frac{\log N}{\log \log N} \right)^{40/19},$$

one has unconditionally

$$\#\{p \leq C_{s,q}Y : p^q \mid \mathcal{F}_s(N)\} \gg_{s,q} \frac{Y}{\log \log N}, \quad (2.34)$$

and

$$\log \prod_{\substack{p \leq C_{s,q}Y \\ p^q \mid \mathcal{F}_s(N)}} p \gg_{s,q} Y. \quad (2.35)$$

For every integer $q \geq 1$, condition (2.33) has an explicit solution. As $q \rightarrow \infty$, its height is $s = 2q + O(\sqrt{q})$. More precisely, put

$$B = \log \frac{1000}{21}, \quad u_q = \left\lfloor B + \sqrt{B^2 + 4Bq} \right\rfloor + 1, \quad s_q = 2q + u_q. \quad (2.36)$$

Then $s_q = 2q + 2\sqrt{Bq} + O(1)$ and $s = s_q$ satisfies (2.33). Thus the necessary height condition $D \geq 2q$ in (2.20) is attained by this explicit family to within $O(\sqrt{q})$. This is a uniform sufficient choice and need not be minimal for an individual value of q .

For $q = 1$, the smallest qualifying member of the dyadic family is

$$F_6(N) = \mathcal{F}_6(N) = \frac{(64N)!}{(N!)^2(2N)!(4N)!(8N)!(16N)!(32N)!}, \quad (2.37)$$

and (2.34)–(2.35) become

$$\#\{p \leq CY : p \mid F_6(N)\} \gg \frac{(\log N)^{40/19}}{(\log \log N)^{59/19}}, \quad \log \text{rad}_{\leq CY} F_6(N) \gg Y. \quad (2.38)$$

For $q = 2$, the smallest qualifying member is $\mathcal{F}_9$; hence the same optimal-order conclusions hold for primes whose squares divide $\mathcal{F}_9(N)$.

Proof. The Landau function of (2.32) is

$$\Delta_s(t) = \sum_{r=0}^{s-1} ([2^{r+1}t] - 2[2^r t]).$$

Its summands are, apart from irrelevant endpoint conventions, the first s binary digits of $\{t\}$. Consequently $D = s$, and

$$\zeta_{\mathcal{F}_s,q} = 2^{-s} \sum_{j=0}^{q-1} \binom{s}{j}. \quad (2.39)$$

The BHP estimate (1.1) gives $QS_{21/40}$ with $c_0 = 0.08$, so (2.33) and Theorem 2.2 prove (2.34)–(2.35).

For the uniform choice in (2.36), let X have the binomial distribution $\text{Bin}(s_q, 1/2)$. Its mean is $q + u_q/2$, and Hoeffding's inequality gives

$$\zeta_{\mathcal{F}_{s_q},q} = \Pr(X < q) \leq \exp\left(-\frac{u_q^2}{2s_q}\right) < e^{-B} = \frac{21}{1000}.$$

Finally, the tails 2^{-s} and $(s+1)2^{-s}$ are decreasing in $s \geq 1$. Now $\zeta_{F_6,1} = 1/64 < 21/1000$, whereas $\zeta_{F_5,1} = 1/32 > 21/1000$. One may take $\alpha = 0.1$ in the proof of Theorem 2.2. For $q = 2$,

$$\zeta_{F_9,2} = \frac{10}{512} = \frac{5}{256} < \frac{21}{1000}, \quad \zeta_{F_8,2} = \frac{9}{256} > \frac{21}{1000};$$

here $\alpha = 1/20$ is admissible because

$$\frac{2\zeta_{F_9,2}}{(21/40)(1-1/20)} = \frac{125}{1596} < 0.08.$$

This proves all assertions. □

3 A robust vertical cover in Pascal's triangle

The next statement retains the two-window geometry of the original problem.

Lemma 3.1 (Two-window incompatibility). *Let $R \geq 2$, and suppose that distinct primes p, q , integers $\mu, \nu \geq 1$, and positive real numbers e_1, e_2 satisfy*

$$P = p^\mu = \frac{k}{2} + e_1, \quad Q = q^\nu = \frac{k}{2} + e_2, \quad e_2 > 2e_1, \quad \frac{e_2}{k} \leq \frac{1}{64(R+1)}.$$

Then every integer n with $2k \leq n \leq Rk$ satisfies

$$p \mid \binom{n}{k} \quad \text{or} \quad q \mid \binom{n}{k}.$$

Proof. For a prime power $P = p^\mu$ with $k/2 < P < k$, grouping the lowest μ base- p digits in Lucas' theorem gives

$$p \nmid \binom{n}{k} \implies n \bmod P \geq k \bmod P = k - P = P - 2e_1.$$

Thus, if neither prime divides, there are integers a, b for which

$$n \in I_a = [aP - 2e_1, aP), \quad n \in J_b = [bQ - 2e_2, bQ). \quad (3.1)$$

The smallness assumption gives $P, Q \leq (1/2 + 1/[64(R+1)])k \leq (1/2 + 1/192)k < 0.506k$; from $2k \leq n \leq Rk$ and (3.1), one may take

$$4 \leq a, b \leq 2R + 1. \quad (3.2)$$

If $a = b = t$, then the left endpoint of J_t lies to the right of the right endpoint of I_t , since

$$t(Q - P) - 2e_2 = t(e_2 - e_1) - 2e_2 > 0$$

for $t \geq 4$ and $e_2 > 2e_1$. If $a \neq b$, then

$$\begin{aligned} |aP - bQ| &\geq \frac{k}{2} - (a+b)e_2 \\ &\geq \frac{k}{2} - (4R+2)e_2 > \frac{7k}{16}. \end{aligned} \quad (3.3)$$

On the other hand, an intersection in (3.1) would imply

$$|aP - bQ| < 2(e_1 + e_2) < 4e_2 \leq \frac{k}{16(R+1)} \leq \frac{k}{48},$$

again a contradiction. Hence the two nondivisibility windows cannot meet. □

Theorem 3.2 (Vertical two-block cloud). *Assume QS_ϑ , and fix $R \geq 2$. For every sufficiently large k , there are two disjoint sets of primes $\mathcal{P}_-(k), \mathcal{P}_+(k)$, independent of n , such that*

$$p \leq C_R \left(\frac{\log k}{\log \log k} \right)^{1/(1-\vartheta)} \quad (p \in \mathcal{P}_- \cup \mathcal{P}_+), \quad (3.4)$$

$$|\mathcal{P}_\pm| \gg_R \frac{(\log k)^{1/(1-\vartheta)}}{(\log \log k)^{1/(1-\vartheta)+2}}, \quad \sum_{p \in \mathcal{P}_\pm} \log p \gg_R \frac{(\log k)^{1/(1-\vartheta)}}{(\log \log k)^{1/(1-\vartheta)+1}}, \quad (3.5)$$

and, for every $2k \leq n \leq Rk$, one has the dichotomy

$$\prod_{p \in \mathcal{P}_-} p \mid \binom{n}{k} \quad \text{or} \quad \prod_{p \in \mathcal{P}_+} p \mid \binom{n}{k}. \quad (3.6)$$

With $\vartheta = 21/40$, if

$$y = C_R \left(\frac{\log k}{\log \log k} \right)^{40/19},$$

then uniformly for $2k \leq n \leq Rk$,

$$\omega_y \binom{n}{k} \gg_R \frac{(\log k)^{40/19}}{(\log \log k)^{78/19}}, \quad (3.7)$$

and

$$\log \text{rad}_{\leq y} \binom{n}{k} \gg_R \frac{(\log k)^{40/19}}{(\log \log k)^{59/19}}. \quad (3.8)$$

Consequently, for every fixed $H > 0$,

$$\text{rad}_{\leq y} \binom{n}{k} > n^H \quad (3.9)$$

for all sufficiently large k , uniformly throughout the vertical strip.

Proof. Apply Lemma 1.3 with its real size parameter denoted by $\mathcal{N} = k/2$, with $c = 1$, $V = 12$, and a fixed small $\eta > 0$. Thus $x_m = (k/2)^{1/m}$. Write

$$x_0 = x_{m_0}, \quad \lambda_m = mx_m^{\vartheta-1}, \quad \lambda_0 = m_0 x_0^{\vartheta-1}.$$

From (1.8)–(1.10), uniformly for $m \in \mathcal{M}$,

$$e^{-3\eta} \lambda_0 \leq \lambda_m \leq \lambda_0 \quad (3.10)$$

once k is sufficiently large. Indeed, the upper bound is (1.11), while $m/m_0 = 1 - o(1)$ and $d \log(x_m/x_0) \leq 2\eta + o(1)$. For the last estimate, at the lower endpoint $m = m_0 - J$ one has

$$d \log \frac{x_m}{x_0} = \frac{dA_0 J}{(m_0 - J)m_0} \leq \eta + o(1), \quad A_0 = \log(k/2),$$

so the displayed factor 2η is a uniform safety margin. Fix η so small that $e^{-3\eta} > 9/10$. By (1.11), one may then choose $B_0 = B_R$ so large that

$$10e^{-3\eta} \lambda_0 > 2(e^{4\lambda_0} - 1), \quad \frac{1}{2}(e^{12\lambda_0} - 1) \leq \frac{1}{64(R+1)} \quad (3.11)$$

for all sufficiently large k .

For each $m \in \mathcal{M}$, define

$$\begin{aligned}\mathcal{P}_{-,m} &= \{p \text{ prime} : x_m + 2h_m \leq p \leq x_m + 4h_m\}, \\ \mathcal{P}_{+,m} &= \{p \text{ prime} : x_m + 10h_m \leq p \leq x_m + 12h_m\},\end{aligned}\tag{3.12}$$

and put $\mathcal{P}_{\pm} = \bigcup_{m \in \mathcal{M}} \mathcal{P}_{\pm,m}$. All these component intervals are pairwise disjoint by (1.12). Lemmas 1.1 and 1.3 therefore give (3.4)–(3.5).

It remains to prove that the dichotomy survives across different exponents. Take $p \in \mathcal{P}_{-,a}$ and $q \in \mathcal{P}_{+,b}$, and write

$$e_- = p^a - \frac{k}{2}, \quad e_+ = q^b - \frac{k}{2}.$$

The binomial theorem, $1 + z \leq e^z$, and (3.10)–(3.11) give

$$\frac{e_-}{k/2} \leq e^{4\lambda_a} - 1 \leq e^{4\lambda_0} - 1,\tag{3.13}$$

whereas Bernoulli's inequality gives

$$\frac{e_+}{k/2} \geq 10\lambda_b \geq 10e^{-3\eta}\lambda_0 > 2\frac{e_-}{k/2}.\tag{3.14}$$

Moreover,

$$\frac{e_+}{k} \leq \frac{1}{2}(e^{12\lambda_b} - 1) \leq \frac{1}{64(R+1)}.\tag{3.15}$$

Thus every global low–high cross-pair satisfies Lemma 3.1, even when $a \neq b$. Consequently a fixed $n \in [2k, Rk]$ cannot have both a nondividing prime in $\mathcal{P}_-$ and one in $\mathcal{P}_+$. If all primes in $\mathcal{P}_-$ divide, take the first alternative in (3.6); otherwise one nondividing low prime forces every prime in $\mathcal{P}_+$ to divide. This proves (3.6), and (1.13) gives (3.7)–(3.8). Finally, the right side of (3.8) is $\omega(\log k)$, while $\log n = \log k + O_R(1)$, so (3.9) follows uniformly. $\square$

Theorem 3.3 (Uniformly expanding vertical strips). *Assume QS_{ϑ} , put $d = 1 - \vartheta$, and let $R = R(k) \geq 2$. Define*

$$L_k = \log(k/2), \quad S = R + 1, \quad U = \log(SL_k), \quad Y = \left(\frac{SL_k}{U}\right)^{1/d}.\tag{3.16}$$

Suppose

$$\frac{U^2}{L_k} \longrightarrow 0,\tag{3.17}$$

equivalently $\log R = o(\sqrt{\log k})$. There are constants $c, C > 0$, depending only on the data in QS_{ϑ} , such that for every sufficiently large k there are disjoint prime sets $\mathcal{P}_-, \mathcal{P}_+ \subset [2, CY]$ satisfying

$$|\mathcal{P}_{\pm}| \geq c \frac{Y}{SU^2}, \quad \sum_{p \in \mathcal{P}_{\pm}} \log p \geq c \frac{Y}{SU},\tag{3.18}$$

and, for every integer $2k \leq n \leq Rk$,

$$\prod_{p \in \mathcal{P}_-} p \mid \binom{n}{k} \quad \text{or} \quad \prod_{p \in \mathcal{P}_+} p \mid \binom{n}{k}.\tag{3.19}$$

Consequently the analogous lower bounds hold for $\omega_{CY}(\frac{n}{k})$ and $\log \text{rad}_{\leq CY}(\frac{n}{k})$, and the latter radical exceeds n^H uniformly for every fixed $H > 0$. In the BHP case,

$$Y = \left(\frac{SL_k}{U}\right)^{40/19}, \quad |\mathcal{P}_{\pm}| \gg \frac{S^{21/19} L_k^{40/19}}{U^{78/19}}, \quad \sum_{p \in \mathcal{P}_{\pm}} \log p \gg \frac{S^{21/19} L_k^{40/19}}{U^{59/19}}. \quad (3.20)$$

The constants c, C are uniform over all admissible functions $R(k)$. The threshold implicit in “sufficiently large” may depend on the chosen function through its rate of convergence in (3.17).

Proof. Choose a fixed $\eta > 0$ so small that $10e^{-3\eta} > 8$. Next choose a fixed sufficiently large B , and put

$$T = U - \log U + B, \quad m_0 = \left\lfloor \frac{dL_k}{T} \right\rfloor, \quad J = \left\lfloor \eta \frac{m_0}{T} \right\rfloor. \quad (3.21)$$

For $m_0 - J \leq m \leq m_0$, set

$$x_m = (k/2)^{1/m}, \quad h_m = x_m^{\vartheta}, \quad \lambda_m = mx_m^{-d},$$

and write $x_0 = x_{m_0}$, $\lambda_0 = \lambda_{m_0}$. Since $m_0 = dL_k/T + O(1)$ and $T \asymp U$, the floor in equation (3.21) gives

$$\frac{L_k}{m_0} = \frac{T}{d} + O\left(\frac{U^2}{L_k}\right).$$

Here the intermediate floor error is $O(T^2/L_k) = O(U^2/L_k)$ because $T \asymp U$. Using (3.17), we therefore obtain, uniformly over the selected exponents,

$$\begin{aligned} x_0^d &= e^B \frac{SL_k}{U} (1 + o(1)), & x_m &\asymp Y, & h_m &\asymp \frac{YU}{SL_k}, \\ \log x_m &\asymp U, & J &\asymp \frac{L_k}{U^2}, & \lambda_0 &= \frac{de^{-B}}{S} (1 + o(1)), \end{aligned} \quad (3.22)$$

Here all $\asymp$ -constants are independent of S , and every $o(1)$ tends to zero along each admissible function $R(k)$, uniformly in the selected exponents. Indeed, $J/m_0 \leq \eta/T$ and the bound $0 \leq \log(x_m/x_0) = L_k(m_0 - m)/(mm_0) \leq \eta/d + o(1)$ holds. In particular,

$$S\lambda_0 = de^{-B} \left(1 + O\left(\frac{\log U}{U} + \frac{U^2}{L_k}\right)\right). \quad (3.23)$$

Moreover,

$$e^{-3\eta}\lambda_0 \leq \lambda_m \leq \lambda_0. \quad (3.24)$$

The consecutive-root calculation from (1.14) now gives

$$\frac{x_{m-1} - x_m}{h_m} \gg SU \longrightarrow \infty. \quad (3.25)$$

Thus all intervals $[x_m, x_m + 12h_m]$ are pairwise disjoint. Define the low and high blocks by (3.12), taking the union over the present exponent set. Lemma 1.1 and (3.22) give

$$|\mathcal{P}_{\pm}| \gg J \frac{h_m}{\log x_m} \asymp \frac{Y}{SU^2}, \quad \sum_{p \in \mathcal{P}_{\pm}} \log p \gg Jh_m \asymp \frac{Y}{SU},$$

with the evident interpretation that any representative h_m may be used because these quantities are uniformly comparable.

Increase B , if necessary. Equations (3.22)–(3.24), uniformly along every admissible choice of $S = S(k) \geq 3$, then imply

$$10e^{-3\eta}\lambda_0 > 2(e^{4\lambda_0} - 1), \quad \frac{1}{2}(e^{12\lambda_0} - 1) \leq \frac{1}{64S}. \quad (3.26)$$

The defect estimates (3.13)–(3.15) now apply verbatim to every low–high cross-pair and verify the hypotheses of Lemma 3.1 with its parameter R , since $S = R + 1$. This proves (3.19), while (3.18) and (3.20) follow from the preceding estimates.

Finally, $\log n \sim L_k$ uniformly in the present range, and

$$\frac{Y/(SU)}{\log n} \asymp \frac{(SL_k)^{\vartheta/d}}{U^{1/d+1}} \longrightarrow \infty.$$

This proves the radical assertion. Notice also that $\log Y = O(U) = o(L_k)$, so the supporting primes remain $k^{o(1)}$. $\square$

4 Short-interval transference: an architecture discussion

This section formulates how the preceding constructions transfer under alternative short-interval hypotheses and records a limitation of the bounded-band architecture. Statements involving hypothetical prime-distribution inputs are explicitly conditional. The preceding proofs use only local prime information and therefore admit a general transfer formulation. Let G be eventually positive with $G(x) = o(x)$, and assume that, for every fixed $C > 1$,

$$G(tx) \asymp_C G(x) \quad \text{uniformly for } C^{-1} \leq t \leq C.$$

Distinguish two inputs:

- an *existence input*, under which every fixed translate of an interval of length greater than $G(x)$ contains a prime;
- a *density input*, under which each such interval contains $\gg G(x)/\log x$ primes.

At a prospective prime scale x , one has $m \asymp \log k / \log x$. The prime-power construction works once

$$\frac{\log k}{\log x} \frac{G(x)}{x} \ll 1,$$

or equivalently

$$\frac{x \log x}{G(x)} \gg \log k. \quad (4.1)$$

The implied constant in (4.1) depends on the Landau plateau or on R . Rounding m to an integer is harmless provided $(\log x)^2 = o(\log k)$, as it is in every row below.

Multiplexing

$$J \asymp \frac{\log k}{(\log x)^2} \quad (4.2)$$

nearby exponents gives disjoint prime bands. The existence input therefore gives at least $\gg J$ distinct divisors with logarithmic mass $\gg J \log x$. Under the density input, the stronger outputs are

$$\#\mathcal{P} \gg \frac{\log k G(x)}{(\log x)^3}, \quad \sum_{p \in \mathcal{P}} \log p \gg \frac{\log k G(x)}{(\log x)^2}. \quad (4.3)$$

Thus a density input yields a radical larger than every fixed power of k when $G(x)/(\log x)^2 \rightarrow \infty$. This extra condition should not be conflated with (4.1), which controls only the stability of the prime powers.

Some benchmark interval scales are shown in Table 1.

Interval scale $G(x)$	Resulting cutoff	Status for a cloud
$x^{21/40}$	$(\log k / \log \log k)^{40/19}$	Published BHP density bound
$x^{13/25}$	$(\log k / \log \log k)^{25/12}$	Existence only in Runbo Li's preprint [3]
$x^{13/25+\varepsilon}$	$(\log k / \log \log k)^{1/(12/25-\varepsilon)}$	Density bound in that preprint, for fixed $\varepsilon > 0$
$\sqrt{x}$	$(\log k / \log \log k)^2$	Hypothetical density input
$\sqrt{x}(\log x)^2$	$O((\log k \log \log k)^2)$	Follows from RH with a sufficiently large constant multiple
$(\log x)^2$	$O(\log k \log \log k)$	Hypothetical full-density input at Cramér length

Table 1: Benchmark short-interval inputs and transferred prime cutoffs.

Runbo Li's Theorem 2 also gives explicit lower-bound constants, but its interval length is $x^{\theta+\varepsilon}$. Thus the quoted $LB(0.525) > 0.249$ does *not* give $QS_{0.525}$ with constant 0.249. Assuming the stated lower bound in that preprint, taking $\theta = 0.524$, $\varepsilon = 0.001$ gives $QS_{0.525}$ with, for example, $c_0 = 0.20$; this makes the five-level dyadic ratio satisfy the dense criterion at the exact BHP cutoff. Taking $\theta = 0.525$ instead makes the four-level ratio available only at cutoff exponent $1/(0.475 - \varepsilon) = 40/19 + O(\varepsilon)$, not at exact exponent $40/19$.

The usual RH prime-gap consequence at length $\sqrt{x} \log x$ gives the existence version at cutoff $O((\log k)^2)$; Cramér's maximum-gap conjecture gives it at $O(\log k \log \log k)$. Neither supplies the local count in (4.3). Even the much stronger full-density statement at Cramér length leaves a factor $\log \log k$ above an $O(\log k)$ cutoff, and at the critical scale its guaranteed logarithmic mass is only of order $\log k$, not superpolynomial.

4.1 What the direct plateau method saturates

Here is a precise architecture-level barrier. Fix constants $C, B > 0$, restrict the roots $k^{1/m}$ to one multiplicative shell $[x/C, Cx]$, and permit at each exponent at most B prime bands of total length $O_B(x^\vartheta)$. The number of eligible exponents is

$$O_C \left(1 + \frac{\log k}{(\log x)^2} \right),$$

including nonconsecutive choices. Brun–Titchmarsh therefore bounds the total number of primes in every such bounded-band certificate by

$$O_{B,C} \left(\left(1 + \frac{\log k}{(\log x)^2} \right) \frac{x^\vartheta}{\log x} \right).$$

At the balanced scale $x = (\log k / \log \log k)^{1/(1-\vartheta)}$, this becomes

$$O_{B,C} \left(\frac{\log k}{(\log \log k)^2} \frac{x^\vartheta}{\log x} \right) = O_{B,C} \left(\frac{x}{(\log \log k)^2} \right).$$

This matches Theorems 2.1 and 3.2 in order. Thus, within the explicitly defined bounded-band architecture, more isolated plateaux or nonconsecutive exponents can improve constants but not the exponent of $\log \log k$. Theorem 2.2 escapes the barrier by using a root interval of width $\asymp x^\theta \log x$ and the oscillation of the full periodic Landau function, rather than a bounded collection of short plateau bands.

5 Position among nearby results

To the best of the author's knowledge, no previous theorem combines pointwise validity for every sufficiently large parameter, general balanced integral factorial ratios, growing prime support below a prescribed polylogarithmic cutoff, fixed prime-power multiplicity, and a quantitative squarefree logarithmic mass. The comparisons below distinguish the present statements from the closest small-prime, factorial-ratio, least-prime-factor, and row-cover results.

Erdős, Graham, Ruzsa and Straus [8] proved average and almost-all laws for small prime factors of the central binomial coefficient. They also recorded the pointwise estimate

$$\sum_{\substack{p \mid \binom{2N}{N} \\ p \leq N}} \frac{1}{p} \gg \log \log N.$$

Thus pointwise abundance of prime divisors is classical. Their cutoff is the natural scale N , however, and the result concerns the central binomial coefficient; it does not give polylogarithmic localization, a fixed multiplicity, or the radical-mass bounds proved here for general ratios.

Sander [9] proved that every sufficiently large near-central binomial coefficient has a prescribed power of some growing prime as a divisor, and extended pointwise prescribed-power divisibility to multinomial coefficients [10]. Kolesnik [11] proved exact p^d -divisibility for many large primes in multinomial settings. These are close in their pointwise and valuation aspects, but they force one or many primes in polynomial or large-prime ranges rather than a polylogarithmically supported cloud with a squarefree mass estimate.

Erdős stated the asymptotic for the total number of distinct prime divisors of fixed-ratio binomial coefficients; Xylouris [12] supplied an explicit proof and a sharper prime-interval identity. Those primes range over all sizes. This does not imply (3.7)–(3.9), where every counted prime is polylogarithmic and the conclusion is pointwise throughout a vertical ratio strip.

Konyagin's least-prime-factor estimate [13] forces one small divisor. Theorem 3.2 instead gives two predetermined, growing blocks, one of which divides in its entirety. This block dichotomy is not a consequence of a bound for the least prime factor.

Theorem 3.2 is a vertical analogue of the row-cover question of Shareshian–Woodroffe [16] and Casacuberta [17]. Their quantifiers fix a row n and seek a few primes covering all k ; here k is fixed and two whole prime blocks cover all $n \in [2k, Rk]$. Neither result contains the other.

Recent work of Alexeev, Putterman, Sawhney, Sellke and Valiant [14] finds, for each row n , some $k \leq (24/(\pi^2 - 6) + o(1))(\log n)^2$ whose prime-at-most- k part, counted with multiplicity, exceeds n^2 . Their prime cutoff is asymptotically smaller than the BHP cutoff in this note. The tradeoff is in the quantifiers and the conclusion: here one fixes k , covers every n in a ratio strip by predetermined blocks, controls the squarefree radical, and obtains every fixed power n^H . Neither theorem contains the other. Eric Li [19] proves a density-one refinement for the row problem. To state a complementary worst-case lower bound from a very recent preprint of Bae [20], put

$$u(n, k) = \prod_{p \leq k} p^{v_p(\binom{n}{k})}, \quad f(n) = \min\{1 \leq k \leq n : u(n, k) > n^2\},$$

with the minimum interpreted as $+\infty$ when the set is empty. Bae proves

$$\limsup_{n \rightarrow \infty} \frac{f(n)}{\log n} \frac{\log \log \log n}{\log \log n} \geq \frac{1}{2}.$$

Pomerance [18] proves density-one divisibility families for the middle binomial coefficient, including uniform divisibility by $\binom{m+k}{k}$ for k in a subpolynomial range. That result is again distributional and specific to $\binom{2m}{m}$, whereas the present factorial-ratio theorems hold for every sufficiently large argument.

Landau [4] and Soundararajan [5, 6] use Δ_F to study integrality and the classification of factorial ratios, including irreducible ratios of height greater than one. Bell–Bober [7] prove the same reflection symmetry used in (2.11), together with mean-square bounds and length-versus-height restrictions for these step functions. Their L^2 bounds also constrain how small the phase measures $\zeta_{F,q}$ can be. Existing Catalan-type results, such as [15], study fixed-prime divisibility or sequence-specific squarefreeness. Theorems 2.1–2.2 instead turn the Landau phases into growing, pointwise sets of polylogarithmic primes with controlled multiplicity.

The results above therefore complement, rather than subsume, the existing factorial-ratio and row-cover literature. Their distinguishing features are the polylogarithmic support for fixed factorial ratios and fixed vertical strips, the subpolynomial support for expanding strips, the pointwise vertical quantifier, and the formulation in terms of general Landau step functions.

Declaration on Generative-AI Assistance

This project was developed with substantial assistance from generative-AI systems, principally OpenAI’s ChatGPT and Codex and Anthropic’s Claude. These systems were used for mathematical exploration, the generation and critique of candidate proof arguments, symbolic and numerical checks, computational experiments and ancillary code, literature-search leads, and the drafting and editing of English and LaTeX. Their outputs were treated as unverified suggestions rather than as mathematical or bibliographic sources. AI-assisted computations are supplementary and are not used as premises in the asymptotic proofs.

Miwa Yuuki directed the project, selected the statements and arguments included here, and takes responsibility for the submitted manuscript and for any remaining errors. The AI systems are not authors.

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